📄 Project Scope Document

EduTrack Academic Management Platform

VTU Internship 2026 — Team 15

1. Project Overview

EduTrack is a full-stack web-based academic management platform designed to digitize and streamline academic operations in educational institutions. It connects three types of users — students, teachers, and administrators — through a centralized platform that manages attendance, assignments, marks, course materials, and system settings in real time.

2. Project Objectives

- Provide a centralized platform for managing academic activities
- Eliminate manual paper-based attendance and marks tracking
- Enable real-time access to academic data for students, teachers, and admins
- Provide secure role-based access for all three user types
- Support OTP-based password recovery for all users
- Allow administrators to control platform-wide settings including maintenance mode and registration

3. Scope

3.1 In Scope

Admin Module

- Login and logout
- Add, edit, delete, and search students
- Add, edit, and delete teachers
- Update student and teacher status
- Reset passwords for students and teachers
- Bulk import students and teachers via CSV

- Enable/disable maintenance mode
- Open/close student self-registration
- Set active academic year and semester

Teacher Module

- Login and logout
- Create, update, and delete courses
- Mark, view, and update student attendance
- View attendance statistics per student
- Enter, update, and view student marks (auto-graded)
- Create, update, and delete assignments
- View student submissions
- Upload and delete course materials
- Update personal profile

Student Module

- Self-registration and login
- View attendance per course with percentage
- View marks and grades per course
- View CGPA
- View and submit assignments
- View and download course materials
- Update personal profile
- Reset password via OTP

Security & System

- BCrypt password hashing for all users
- OTP-based password reset via Gmail SMTP
- Role-based access (separate portals per role)
- Maintenance mode to block student/teacher logins
- CORS support for cross-origin API access

Testing

- 43 Playwright end-to-end tests
- 31 JUnit unit and controller tests

- 26 Postman API requests

Deployment

- Frontend deployed on Vercel
- Backend deployed on Render (Docker)
- Database hosted on Railway (MySQL)

3.2 Out of Scope

The following features are **not** included in this version:

- Mobile application (Android/iOS)
- Real-time notifications or push alerts
- Video conferencing or live classes
- Fee management or payment processing
- Library management
- Timetable or scheduling system
- Parent portal
- Third-party authentication (Google/Facebook login)
- Advanced analytics or reporting dashboards
- Multi-institution support

4. Deliverables

DELIVERABLE	DESCRIPTION
Frontend Application	React 19 web app deployed on Vercel
Backend REST API	Spring Boot 3 API deployed on Render
Database	MySQL schema with 11 tables on Railway
Source Code	GitHub repository with full codebase
Test Suite	43 Playwright + 31 JUnit tests
Postman Collection	26 API requests for all endpoints
System Design Document	Architecture, DB design, API overview
User Stories	64 user stories across 3 roles
WBS	Work breakdown structure with task assignments

DELIVERABLE	DESCRIPTION
Project Scope Document	This document

5. Assumptions

- Users have a stable internet connection to access the platform
- Gmail SMTP is available for OTP email delivery
- MySQL 8+ is used as the database
- The platform is used by a single institution
- Admin accounts are created manually (no self-registration for admins)
- Department and academic details (year, semester) cannot be changed after registration

6. Constraints

CONSTRAINT	DETAILS
Technology	React, Spring Boot, MySQL only
Deployment	Free tier hosting (Render sleeps after 15 min inactivity)
Time	7-week internship timeline
Team Size	9 members
Authentication	No JWT — session-less custom auth

7. Risks

RISK	IMPACT	MITIGATION
Backend sleep on free tier	Slow first request after inactivity	Wake up backend before demo
Gmail SMTP limits	OTP emails may be delayed	Use Gmail App Password, not main password
Database connection loss	App becomes unavailable	Railway MySQL is managed and reliable
Scope creep	Delayed delivery	Strict scope defined in this document

8. Stakeholders

STAKEHOLDER	ROLE
VTU Internship Evaluators	Project reviewers
Naveen P R	Team Leader & Core Developer
Team Members (8)	Developers
Students (end users)	Use student portal
Teachers (end users)	Use teacher portal
Admin (end user)	Use admin portal

9. Timeline

PHASE	DURATION
Planning & Requirements	Week 1
Database Design	Week 1–2
Backend Development	Week 2–4
Frontend Development	Week 3–5
Integration & Bug Fixes	Week 5–6
Testing	Week 6
Deployment	Week 7

10. Success Criteria

- All three portals (Admin, Teacher, Student) are fully functional
- OTP-based password reset works end to end
- All 43 Playwright tests pass
- All 31 JUnit tests pass
- Application is accessible via public URL
- Admin can manage students, teachers, and system settings
- Teachers can mark attendance and enter marks

- Students can view their academic data